

A fully worked finite-horizon likelihood example for a random-lifetime frog model: $K = 5$, $T = 4$, and $\beta_0 = 0.45$

Standalone numerical illustration

Abstract. This document gives a fully specified numerical realization of the labeled finite-horizon experiment for a one-dimensional frog model with beta-geometric particle lifetimes. We fix $K = 5$ independent realizations, observation horizon $T = 4$, and generating parameter $\beta_0 = 0.45$. The purpose is to make the notation operational by displaying the activation times, potential paths, observed states and positions, at-risk and death indicators, right-censoring variables, global activation order, age-specific counts, two elementary estimators, and the exact likelihood for one concrete record. Latent quantities are reported because they are available in simulation; in the statistical experiment itself, only the labeled state-position record through time T is observed. The document is intended as a self-contained repository note rather than as part of a journal manuscript.

1. Simulation design

We consider one initially dormant particle at each site of $\mathbb{Z}$ and one initially active particle at the origin of each realization. A particle is indexed by (r, x) , where $r \in \{1, \dots, 5\}$ denotes the independent realization and $x \in \mathbb{Z}$ denotes the particle's home site.

For each label (r, x) , generate a latent survival mark

$$\Pi_{r,x} \sim \text{Beta}(1, 0.45).$$

Conditional on $\Pi_{r,x} = u$, the potential lifetime $\Xi_{r,x}$ has mass function

$$\mathbb{P}(\Xi_{r,x} = j \mid \Pi_{r,x} = u) = u^j(1 - u), \quad j \in \mathbb{N}_0.$$

Thus, $\Xi_{r,x}$ is the total number of jumps that the particle would complete after activation and before death. Independently of all lifetime variables, the particle receives an i.i.d. direction sequence $\varepsilon_{r,x}(m) \in \{-1, +1\}$ with equal probabilities.

The simulation uses random seed 9. For reproducibility, the latent variables are generated from independent uniforms U_1, U_2 by

$$\Pi = 1 - (1 - U_1)^{1/\beta_0}, \quad \Xi = \left\lfloor \frac{\log U_2}{\log \Pi} \right\rfloor,$$

followed by symmetric directions. The value $\beta_0 = 0.45$ is used only to generate the record. In the likelihood and estimation calculations below, β is again treated as unknown.

Remark 1.1. *The quantities $\Pi_{r,x}$, $\Xi_{r,x}$, and all unobserved future directions are latent. They are displayed here only because this is a simulated example. In an actual labeled experiment, the observed data are the particle states and positions at calendar times 0, 1, 2, 3, 4.*

2. Dictionary of variables

The following notation is used for an activated particle (r, x) .

$A_{r,x}$: The calendar time at which the particle is activated. It is dormant before $A_{r,x}$ and active at time $A_{r,x}$.

$X_{r,x}(m)$: The potential position after m completed jumps:

$$X_{r,x}(0) = x, \quad X_{r,x}(m) = x + \sum_{i=1}^m \varepsilon_{r,x}(i).$$

$H_{r,x}(t)$: The particle state at calendar time t : dormant (sl), active and alive (al), or dead (de).

$L_{r,x}(t)$: The recorded position at time t . Before activation it equals the home site; after death it remains equal to the last occupied site.

$C_{r,x}(T)$: The number of transition opportunities before the horizon:

$$C_{r,x}(T) = T - A_{r,x} \quad \text{when } A_{r,x} < T,$$

and $C_{r,x}(T) = 0$ when $A_{r,x} = T$.

$Z_{r,x}(T)$: The observed lifetime length,

$$Z_{r,x}(T) = \min\{\Xi_{r,x}, C_{r,x}(T)\}.$$

It is the number of jumps observed before death or before the observation horizon, whichever occurs first.

$\Delta_{r,x}(T)$: The observed-death indicator,

$$\Delta_{r,x}(T) = \mathbf{1}_{\{\Xi_{r,x} < C_{r,x}(T)\}}.$$

If $\Delta_{r,x}(T) = 0$, the lifetime is right-censored at the horizon.

$Y_{r,x,j}(T)$: The indicator that the age- j transition is visible:

$$Y_{r,x,j}(T) = \mathbf{1}_{\{A_{r,x}+j < T, \Xi_{r,x} \geq j\}}.$$

$D_{r,x,j}(T)$: The indicator that a visible age- j transition ends in death:

$$D_{r,x,j}(T) = \mathbf{1}_{\{A_{r,x}+j < T, \Xi_{r,x} = j\}}.$$

$J_{r,x,j}(T)$: The visible-jump indicator,

$$J_{r,x,j}(T) = Y_{r,x,j}(T) - D_{r,x,j}(T).$$

$Z_n^{(5)}$: The n th particle in the global activation order across the five realizations. This notation must not be confused with $Z_{r,x}(T)$, the observed lifetime length of one particle.

3. Labeled trajectories and activation times

Figure 1 displays every activated particle. An open circle marks activation, a cross marks an observed death, and an open square marks a particle that is still alive when observation stops at $T = 4$.

Labeled trajectories through $T = 4$: circle = activation, x = observed death, square = right censoring

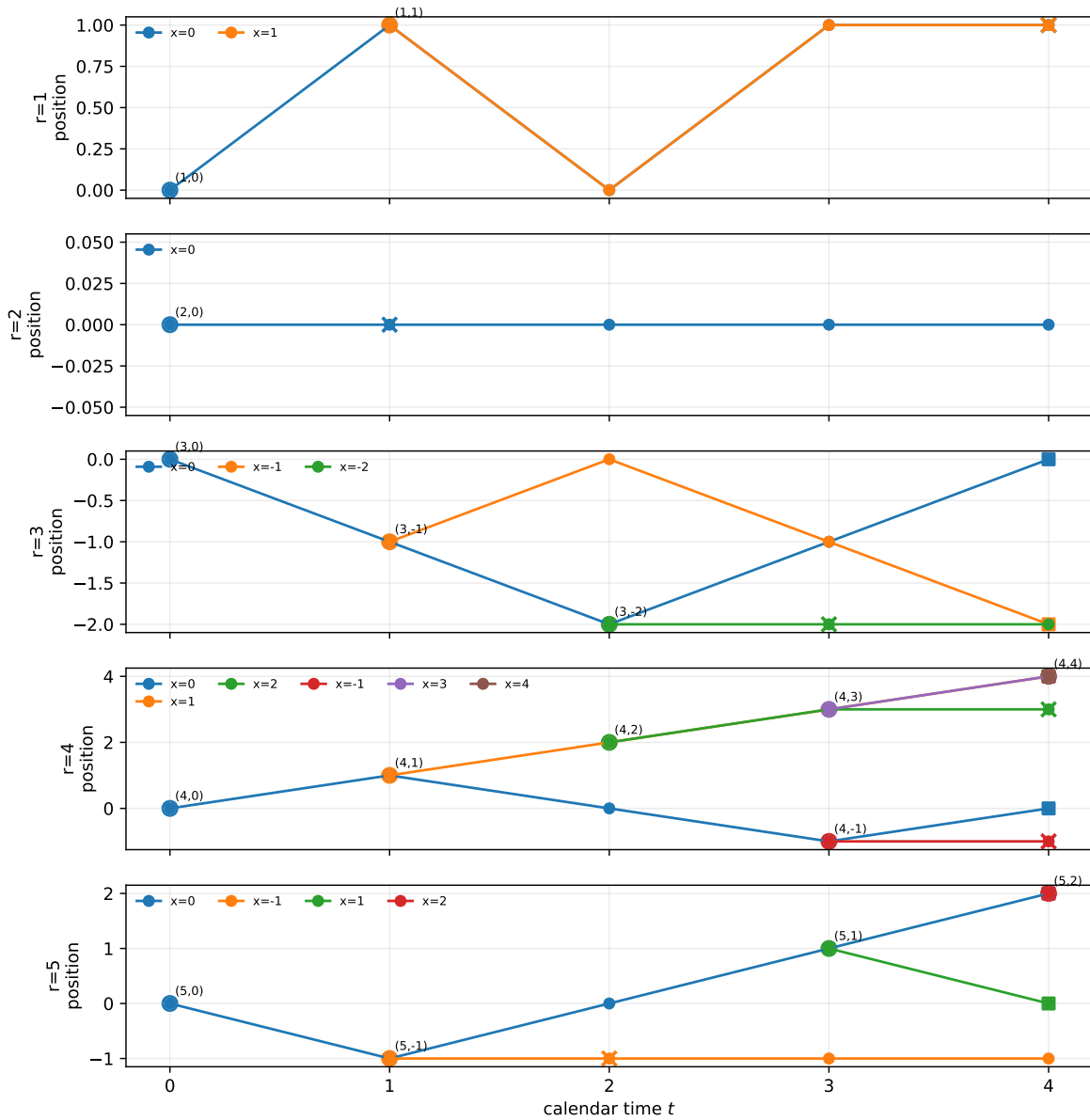


FIGURE 1. Labeled particle trajectories through the finite horizon.

The five realizations can be summarized as follows.

- Realization 1.** Particle $(1,0)$ is activated at time 0, completes three jumps, and its death is observed. Particle $(1,1)$ is activated at time 1, completes two jumps, and its death is observed.
- Realization 2.** Particle $(2,0)$ is activated at time 0, completes no jump, and its death is observed.
- Realization 3.** Particle $(3,0)$ is activated at time 0, completes four observed jumps, and is right-censored. Particle $(3,-1)$ is activated at time 1, completes three observed jumps, and is right-censored. Particle $(3,-2)$ is activated at time 2, completes no jump, and its death is observed.

- Realization 4.** Particle $(4,0)$ is activated at time 0, completes four observed jumps, and is right-censored. Particle $(4,1)$ is activated at time 1, completes three observed jumps, and is right-censored. Particle $(4,2)$ is activated at time 2, completes one jump, and its death is observed. Particle $(4,-1)$ is activated at time 3, completes no jump, and its death is observed. Particle $(4,3)$ is activated at time 3, completes one observed jump, and is right-censored. Particle $(4,4)$ is activated exactly at time T and therefore contributes no transition of its own.
- Realization 5.** Particle $(5,0)$ is activated at time 0, completes four observed jumps, and is right-censored. Particle $(5,-1)$ is activated at time 1, completes no jump, and its death is observed. Particle $(5,1)$ is activated at time 3, completes one observed jump, and is right-censored. Particle $(5,2)$ is activated exactly at time T and therefore contributes no transition of its own.

3.1. *Global activation order.* The sequence $Z_n^{(5)}$ begins with the five origin particles. All subsequent particles are ordered by activation time, with deterministic tie-breaking. [Figure 2](#) gives a graphical representation and [Table 1](#) reports the exact values.

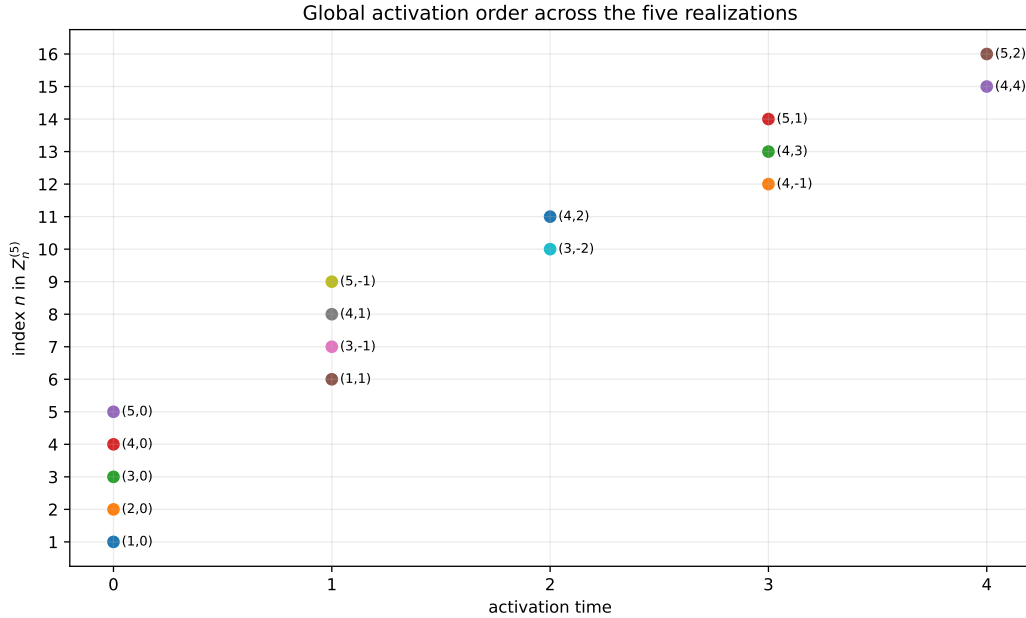


FIGURE 2. Global activation index n and activation time of $Z_n^{(5)}$.

TABLE 1. Activation order and associated latent values. Survival marks are displayed to twelve decimal places.

n	$Z_n^{(5)} = (r, x)$	$A_{r,x}$	$\Pi_{r,x}$	$\Xi_{r,x}$
1	(1, 0)	0	0.748852052343	3
2	(2, 0)	0	0.658864800511	0
3	(3, 0)	0	0.867178100529	14
4	(4, 0)	0	0.874058328918	14
5	(5, 0)	0	0.850211770952	24
6	(1, 1)	1	0.679706918303	2
7	(3, -1)	1	0.840255727892	11
8	(4, 1)	1	0.965480621624	42
9	(5, -1)	1	0.547795716564	0
10	(3, -2)	2	0.146829468108	0
11	(4, 2)	2	0.377623703115	1
12	(4, -1)	3	0.474265696617	0
13	(4, 3)	3	0.677549874181	4
14	(5, 1)	3	0.947083752203	28
15	(4, 4)	4	0.978962476308	0
16	(5, 2)	4	0.995044411438	56

4. Particle-level values

Table 2 collects the principal latent and observed variables. The column $X(0:Z)$ gives the observed segment of the potential path. For a particle activated exactly at T , one has $C = Z = 0$, $\Delta = 0$, and the particle's own survival contribution is $S_\beta(0) = 1$.

TABLE 2. Latent and observed values for every activated particle.

n	(r, x)	A	Π	Ξ	C	Z	Δ	$X(0:Z)$	status at T
1	(1, 0)	0	0.748852052343	3	4	3	1	(0, 1, 0, 1)	dead
2	(2, 0)	0	0.658864800511	0	4	0	1	(0)	dead
3	(3, 0)	0	0.867178100529	14	4	4	0	(0, -1, -2, -1, 0)	censored
4	(4, 0)	0	0.874058328918	14	4	4	0	(0, 1, 0, -1, 0)	censored
5	(5, 0)	0	0.850211770952	24	4	4	0	(0, -1, 0, 1, 2)	censored
6	(1, 1)	1	0.679706918303	2	3	2	1	(1, 0, 1)	dead
7	(3, -1)	1	0.840255727892	11	3	3	0	(-1, 0, -1, -2)	censored
8	(4, 1)	1	0.965480621624	42	3	3	0	(1, 2, 3, 4)	censored
9	(5, -1)	1	0.547795716564	0	3	0	1	(-1)	dead
10	(3, -2)	2	0.146829468108	0	2	0	1	(-2)	dead
11	(4, 2)	2	0.377623703115	1	2	1	1	(2, 3)	dead
12	(4, -1)	3	0.474265696617	0	1	0	1	(-1)	dead
13	(4, 3)	3	0.677549874181	4	1	1	0	(3, 4)	censored
14	(5, 1)	3	0.947083752203	28	1	1	0	(1, 0)	censored
15	(4, 4)	4	0.978962476308	0	0	0	0	(4)	active at T
16	(5, 2)	4	0.995044411438	56	0	0	0	(2)	active at T

Figure 3 displays the distinction between observed death and right censoring. A cross indicates that death occurs inside the observation window; a square indicates that observation stops while the particle remains alive. Particles activated exactly at T have zero exposure after activation.

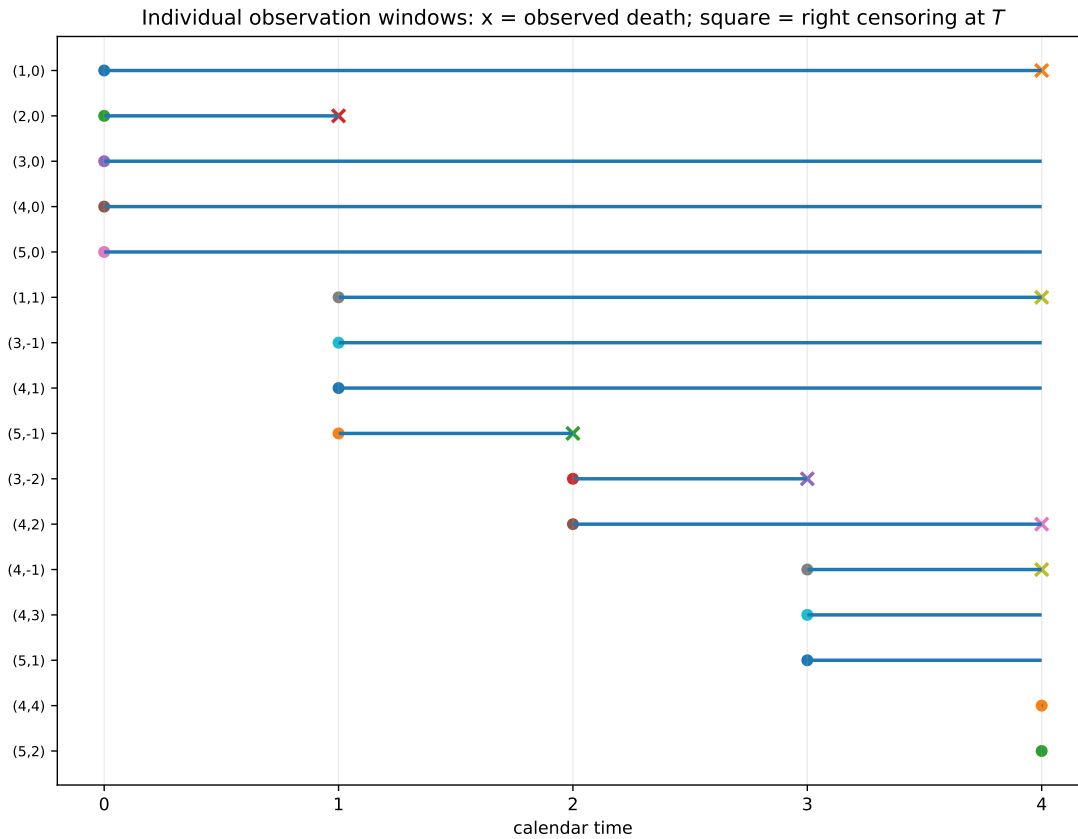


FIGURE 3. Individual observation windows and terminal observation type.

4.1. *Observed state-position record.* In Table 3, an entry $sl(x)$ means that the particle is dormant at its home site; $al(y)$ means that it is active and alive at position y ; and $de(y)$ means that it is dead, with the last occupied position retained for bookkeeping.

TABLE 3. Observed values of $(H_{r,x}(t), L_{r,x}(t))$ at each calendar time.

Particle	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$
(1, 0)	al(0)	al(1)	al(0)	al(1)	de(1)
(2, 0)	al(0)	de(0)	de(0)	de(0)	de(0)
(3, 0)	al(0)	al(-1)	al(-2)	al(-1)	al(0)
(4, 0)	al(0)	al(1)	al(0)	al(-1)	al(0)
(5, 0)	al(0)	al(-1)	al(0)	al(1)	al(2)
(1, 1)	sl(1)	al(1)	al(0)	al(1)	de(1)
(3, -1)	sl(-1)	al(-1)	al(0)	al(-1)	al(-2)
(4, 1)	sl(1)	al(1)	al(2)	al(3)	al(4)
(5, -1)	sl(-1)	al(-1)	de(-1)	de(-1)	de(-1)
(3, -2)	sl(-2)	sl(-2)	al(-2)	de(-2)	de(-2)
(4, 2)	sl(2)	sl(2)	al(2)	al(3)	de(3)
(4, -1)	sl(-1)	sl(-1)	sl(-1)	al(-1)	de(-1)
(4, 3)	sl(3)	sl(3)	sl(3)	al(3)	al(4)
(5, 1)	sl(1)	sl(1)	sl(1)	al(1)	al(0)
(4, 4)	sl(4)	sl(4)	sl(4)	sl(4)	al(4)
(5, 2)	sl(2)	sl(2)	sl(2)	sl(2)	al(2)

4.2. *Particle-age rows.* For each activated particle, Table 4 reports the vectors (Y_0, Y_1, Y_2, Y_3) , (D_0, D_1, D_2, D_3) , and (J_0, J_1, J_2, J_3) . These are exactly the indicators used in the hazard representation of the likelihood.

TABLE 4. At-risk, death, and jump indicators by individual age.

Particle	A	(Y_0, Y_1, Y_2, Y_3)	(D_0, D_1, D_2, D_3)	(J_0, J_1, J_2, J_3)
(1, 0)	0	(1, 1, 1, 1)	(0, 0, 0, 1)	(1, 1, 1, 0)
(2, 0)	0	(1, 0, 0, 0)	(1, 0, 0, 0)	(0, 0, 0, 0)
(3, 0)	0	(1, 1, 1, 1)	(0, 0, 0, 0)	(1, 1, 1, 1)
(4, 0)	0	(1, 1, 1, 1)	(0, 0, 0, 0)	(1, 1, 1, 1)
(5, 0)	0	(1, 1, 1, 1)	(0, 0, 0, 0)	(1, 1, 1, 1)
(1, 1)	1	(1, 1, 1, 0)	(0, 0, 1, 0)	(1, 1, 0, 0)
(3, -1)	1	(1, 1, 1, 0)	(0, 0, 0, 0)	(1, 1, 1, 0)
(4, 1)	1	(1, 1, 1, 0)	(0, 0, 0, 0)	(1, 1, 1, 0)
(5, -1)	1	(1, 0, 0, 0)	(1, 0, 0, 0)	(0, 0, 0, 0)
(3, -2)	2	(1, 0, 0, 0)	(1, 0, 0, 0)	(0, 0, 0, 0)
(4, 2)	2	(1, 1, 0, 0)	(0, 1, 0, 0)	(1, 0, 0, 0)
(4, -1)	3	(1, 0, 0, 0)	(1, 0, 0, 0)	(0, 0, 0, 0)
(4, 3)	3	(1, 0, 0, 0)	(0, 0, 0, 0)	(1, 0, 0, 0)
(5, 1)	3	(1, 0, 0, 0)	(0, 0, 0, 0)	(1, 0, 0, 0)
(4, 4)	4	(0, 0, 0, 0)	(0, 0, 0, 0)	(0, 0, 0, 0)
(5, 2)	4	(0, 0, 0, 0)	(0, 0, 0, 0)	(0, 0, 0, 0)

Summing over particles gives

$$(R(0), R(1), R(2), R(3)) = (14, 8, 7, 4), \quad (d(0), d(1), d(2), d(3)) = (4, 1, 1, 1).$$

The numbers of visible jumps by age are therefore $(10, 7, 6, 3)$, and the total number of observed jumps is

$$G_{5,4} = 10 + 7 + 6 + 3 = 26.$$

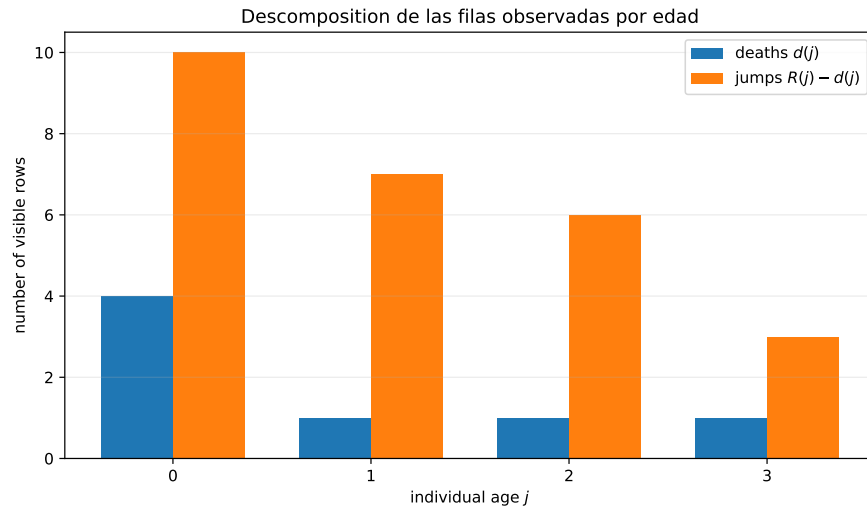


FIGURE 4. Observed deaths and jumps by individual age.

5. Two elementary estimators

5.1. *Estimator based on the origin particles.* Define

$$I_r = \mathbf{1}_{\{\Xi_{r,0} \geq 1\}}.$$

For the simulated record,

$$(\Xi_{1,0}, \dots, \Xi_{5,0}) = (3, 0, 14, 14, 24), \quad (I_1, \dots, I_5) = (1, 0, 1, 1, 1).$$

Hence

$$\hat{q}_{\text{root}} = \frac{4}{5}, \quad \hat{\beta}_{\text{root}} = \hat{q}_{\text{root}}^{-1} - 1 = \frac{1}{4} = 0.25.$$

This estimator uses one guaranteed first transition from each independent realization.

5.2. *Pooled estimator based on visible first transitions.* There are

$$N_{5,4} = 14$$

particles activated strictly before the horizon and therefore 14 visible first transitions. Among them, $M_{5,4} = 10$ end in a jump. Consequently,

$$\hat{q}_{\text{act}} = \frac{10}{14} = \frac{5}{7}, \quad \hat{\beta}_{\text{act}} = \hat{q}_{\text{act}}^{-1} - 1 = \frac{2}{5} = 0.4.$$

Although this ratio resembles an ordinary binomial estimator, its justification is not based on a conditional binomial law for $M_{5,4}$ given $N_{5,4}$. The number and identities of included particles are generated by the process itself; the theoretical argument uses predictable revelation and a martingale estimating identity.

6. Exact likelihood for the simulated record

The survival function and age-specific death hazard are

$$S_\beta(j) = \mathbb{P}_\beta(\Xi \geq j), \quad h_\beta(j) = \mathbb{P}_\beta(\Xi = j \mid \Xi \geq j) = \frac{\beta}{\beta + j + 1},$$

and $p_\beta(j) = S_\beta(j)h_\beta(j)$.

6.1. *Particle-level censored-lifetime representation.* Observed deaths occur for

$$(1, 0), (2, 0), (5, -1), (3, -2), (4, -1), (4, 2), (1, 1),$$

with observed lifetimes 3, 0, 0, 0, 0, 1, 2, respectively. The right-censored particles activated before T contribute three factors $S_\beta(4)$, two factors $S_\beta(3)$, and two factors $S_\beta(1)$. The two particles activated exactly at T contribute $S_\beta(0) = 1$. Therefore,

$$\boxed{L_{5,4}(\beta) = 2^{-26} p_\beta(0)^4 p_\beta(1) p_\beta(2) p_\beta(3) S_\beta(4)^3 S_\beta(3)^2 S_\beta(1)^2.} \quad (1)$$

The factor 2^{-26} is the probability of the 26 observed directions. No separate activation factor is required: once previous transitions and directions are fixed, all activations in the labeled record are deterministic.

6.2. *Age-compressed hazard representation.* The same likelihood can be written as

$$L_{5,4}(\beta) = 2^{-26} \prod_{j=0}^3 h_{\beta}(j)^{d(j)} \{1 - h_{\beta}(j)\}^{R(j)-d(j)}.$$

Substituting the observed counts gives

$$\boxed{L_{5,4}(\beta) = 2^{-26} \left(\frac{\beta}{\beta+1} \right)^4 \left(\frac{1}{\beta+1} \right)^{10} \left(\frac{\beta}{\beta+2} \right) \left(\frac{2}{\beta+2} \right)^7} \quad (2)$$

$$\times \left(\frac{\beta}{\beta+3} \right) \left(\frac{3}{\beta+3} \right)^6 \left(\frac{\beta}{\beta+4} \right) \left(\frac{4}{\beta+4} \right)^3.$$

This expression uses all visible information about deaths, positive-age jumps, and terminal censoring.

6.3. *Maximum-likelihood estimate.* Let $\theta = \log \beta$. After removing terms independent of θ , the log-likelihood is

$$\ell_{5,4}(\theta) = 7\theta - 14 \log(e^{\theta} + 1) - 8 \log(e^{\theta} + 2) - 7 \log(e^{\theta} + 3) - 4 \log(e^{\theta} + 4).$$

The score is

$$U_{5,4}(\theta) = 7 - 14h_{\theta}(0) - 8h_{\theta}(1) - 7h_{\theta}(2) - 4h_{\theta}(3).$$

Thus, the score equation $U_{5,4}(\hat{\theta}) = 0$ is equivalent to

$$7 = \frac{14\beta}{\beta+1} + \frac{8\beta}{\beta+2} + \frac{7\beta}{\beta+3} + \frac{4\beta}{\beta+4}.$$

Its positive solution is

$$\hat{\beta}_{\text{MLE}} \approx 0.4390.$$

Figure 5 displays the relative likelihood and compares its maximizer with the generating value $\beta_0 = 0.45$.

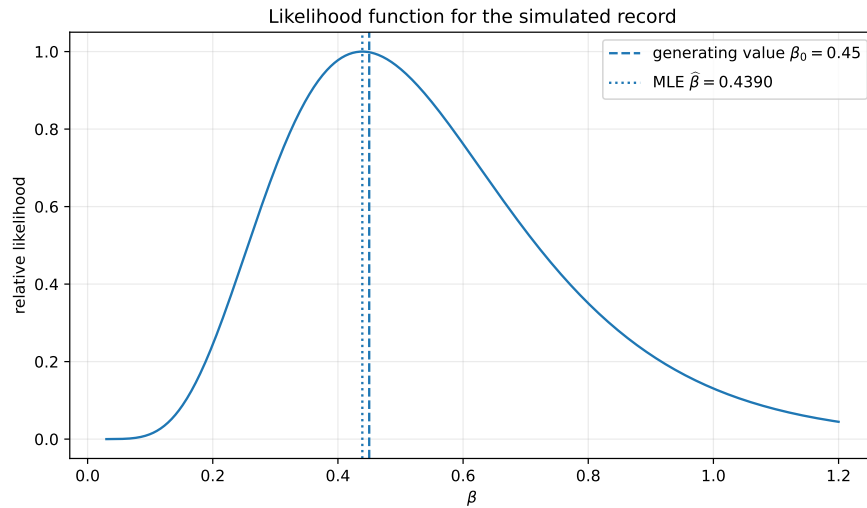


FIGURE 5. Relative likelihood for the simulated labeled record.

7. Summary and interpretation

The generating parameter and the three point estimates are

$$\beta_0 = 0.45, \quad \hat{\beta}_{\text{root}} = 0.25, \quad \hat{\beta}_{\text{act}} = 0.40, \quad \hat{\beta}_{\text{MLE}} \approx 0.4390.$$

With only five independent realizations, the origin estimator uses five binary observations and is consequently the least stable in this example. The pooled estimator uses 14 visible first transitions. The maximum-likelihood estimator additionally uses transitions at ages 1, 2, 3 and the information carried by terminal right censoring, and it lies closest to the generating parameter in this realization.

Since $\beta_0 = 0.45 < 1/2$, the generating value lies in the positive-survival regime of the off-critical phase result. All three point estimates also lie below $1/2$ in this sample. This is only a descriptive observation: formal phase classification requires a confidence set or a hypothesis test rather than the sign of a point estimate alone.

Remark 7.1. *This example distinguishes right censoring from a death observed at the horizon. If a particle is alive at T , the data imply only $\Xi \geq C(T)$ and the likelihood contains $S_\beta(C(T))$. If it is recorded as dead at T , the death occurred during the last observed transition and the exact value of Ξ is known, so the likelihood contains $p_\beta(\Xi)$. A particle activated exactly at T has $C(T) = 0$ and contributes $S_\beta(0) = 1$ through its own lifetime; its activation has already been accounted for by the previous observed jump that reached its home site.*

Reproducibility files

The accompanying repository directory contains this L^AT_EX source, all figures in PDF format, the simulation output in JSON format, and the Python script used to reproduce the record and figures.